

Experiment No. 2

Aim:-

To study and execute basic networking commands in Linux and Windows for checking network configuration, connectivity, and troubleshooting network-related issues.

Requirements:-

- Computer or Laptop
- Windows or Linux Operating System
- Command Prompt (Windows) or Terminal (Linux)
- Active Network Connection (LAN/Wi-Fi)

Theory:-

Computer networking is the process of connecting two or more computers or devices to share data, resources, and services over a communication network. Every device connected to a network is identified by a unique IP (Internet Protocol) address, which enables communication between devices. Network administrators use networking commands to configure, monitor, and troubleshoot network connections efficiently.

Networking commands are built-in command-line utilities available in both Windows (Command Prompt) and Linux (Terminal) operating systems. These commands provide detailed information about the system's network configuration, test communication with remote hosts, identify network faults, and monitor active connections. They are widely used in system administration, network maintenance, and cybersecurity.

Some of the most commonly used networking commands include:

- `ipconfig` (Windows) Displays the current TCP/IP configuration of the computer, including IP address, subnet mask, default gateway, and DNS server information. The `'ipconfig /all'` command provides detailed network configuration.
- `ip addr / ifconfig` (Linux) Displays network interface information such as IP addresses, MAC addresses, interface status, and packet statistics. The `'ip'` command is the modern replacement for `'ifconfig'`.
- `ping`: Sends ICMP (Internet Control Message Protocol) Echo Request packets to a remote host and measures the response time. It is used to verify network connectivity and `tracert` (Windows) / `traceroute` (Linux): Shows the complete path taken by packets from the source computer to the destination host. It helps identify delays or failures at specific routers along the route.
- `nslookup` Queries the Domain Name System (DNS) to translate domain names (such as `'google.com'`) into IP addresses and vice versa. It is useful for checking DNS server functionality.

- netstat Displays active TCP and UDP connections, routing tables, interface statistics, and listening ports. It is commonly used to monitor network activity and detect unauthorized connections.

These networking commands help users diagnose problems such as incorrect IP configuration, DNS resolution failures, packet loss, high network latency, and routing issues. They are essential tools for network engineers, system administrators, and students learning computer networking concepts.

Understanding and practicing these commands improves knowledge of network communication, troubleshooting techniques, and the working of Internet protocols, making them an important part of networking laboratory experiments.

Procedure:-

In Windows

1. Click Start and open Command Prompt (CMD).
2. Execute the following commands one by one:
 - ipconfig
 - ping
 - tracert
 - nslookup
 - pathping
 - arp
 - getmac
 - netstat
3. Observe and note the output of each command.

In Linux

1. Open the Terminal.
2. Execute the following commands:
 - ifconfig
 - ping
 - traceroute
 - nslookup
 - mtr
 - arp
 - ip link show
 - ss
3. Observe and record the output.

Observation:-

Command	Observation
ipconfig	Displayed the system's IP configuration, including the IPv4 address, subnet mask, default gateway, and network adapter details.
ping	Successfully sent ICMP Echo Request packets to the destination and received replies, confirming that the network connection was active. It also displayed the round-trip time and packet loss statistics.
tracert	Displayed the route (hops) taken by packets from the source computer to the destination host, along with the response time for each router.
pathping	Combined the features of ping and tracert . It traced the route to the destination and calculated packet loss and latency at each hop, helping identify network bottlenecks.
arp	Displayed the Address Resolution Protocol (ARP) cache, showing the mapping between IP addresses and their corresponding MAC (physical) addresses on the local network.
netstat	Displayed active TCP and UDP connections, listening ports, network statistics, and routing information, indicating the current network activity of the system.
getmac	Displayed the Media Access Control (MAC) addresses of all available network adapters installed on the computer along with their transport names.
nslookup	Queried the DNS server and successfully resolved the domain name into its corresponding IP address, confirming proper DNS functionality.

Output:-

```
Command Prompt - nslookup

C:\Users\User>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 10.0.1.33
    Subnet Mask . . . . . : 255.0.0.0
    IPv4 Address. . . . . : 172.16.43.123
    Subnet Mask . . . . . : 255.255.192.0
    IPv4 Address. . . . . : 192.168.10.51
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.2.1

C:\Users\User>getmac

Physical Address      Transport Name
-----
C8-1F-66-32-08-3D    \Device\NPF{A773F4B8-867A-41D3-BB24-787AAC8928F9}

C:\Users\User>nslookup
Default Server:  dns.google
Address:  8.8.8.8
>
```

```
Command Prompt

C:\Users\User>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
          [-r count] [-s count] [[-j host-list] | [-k host-list]]
          [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
          [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f            Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R            Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p            Ping a Hyper-V Network Virtualization provider address.
    -4            Force using IPv4.
    -6            Force using IPv6.

C:\Users\User>
```

```
Command Prompt

C:\Users\User>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

-a          Displays current ARP entries by interrogating the current
            protocol data.  If inet_addr is specified, the IP and Physical
            addresses for only the specified computer are displayed.  If
            more than one network interface uses ARP, entries for each ARP
            table are displayed.
-g          Same as -a.
-v          Displays current ARP entries in verbose mode.  All invalid
            entries and entries on the loop-back interface will be shown.
inet_addr   Specifies an internet address.
-N if_addr  Displays the ARP entries for the network interface specified
            by if_addr.
-d          Deletes the host specified by inet_addr.  inet_addr may be
            wildcarded with * to delete all hosts.
-s          Adds the host and associates the Internet address inet_addr
            with the Physical address eth_addr.  The Physical address is
            given as 6 hexadecimal bytes separated by hyphens.  The entry
            is permanent.
eth_addr     Specifies a physical address.
if_addr     If present, this specifies the Internet address of the
            interface whose address translation table should be modified.
            If not present, the first applicable interface will be used.

Example:
> arp -s 157.55.85.212 00-aa-00-62-c6-09 .... Adds a static entry.
> arp -a          .... Displays the arp table.

C:\Users\User>
```

```
Command Prompt

C:\Users\User>tracert

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
              [-R] [-S srcaddr] [-4] [-6] target_name

Options:
-d          Do not resolve addresses to hostnames.
-h maximum_hops  Maximum number of hops to search for target.
-j host-list    Loose source route along host-list (IPv4-only).
-w timeout      Wait timeout milliseconds for each reply.
-R           Trace round-trip path (IPv6-only).
-S srcaddr      Source address to use (IPv6-only).
-4           Force using IPv4.
-6           Force using IPv6.

C:\Users\User>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
               [-p period] [-q num_queries] [-w timeout]
               [-4] [-6] target_name

Options:
-g host-list    Loose source route along host-list.
-h maximum_hops  Maximum number of hops to search for target.
-i address      Use the specified source address.
-n          Do not resolve addresses to hostnames.
-p period      Wait period milliseconds between pings.
-q num_queries  Number of queries per hop.
-w timeout      Wait timeout milliseconds for each reply.
-4           Force using IPv4.
-6           Force using IPv6.

C:\Users\User>
```

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>netstat

Active Connections

Proto Local Address Foreign Address State
TCP 172.16.43.123:54217 4.213.25.242:https ESTABLISHED
TCP 172.16.43.123:54303 172.16.9.32:8009 ESTABLISHED
TCP 172.16.43.123:54378 lcbomp-in-f188:5228 ESTABLISHED
TCP 172.16.43.123:58924 lcboma-aw-in-f5:https ESTABLISHED
TCP 172.16.43.123:58925 fx-in-f102:https ESTABLISHED
TCP 172.16.43.123:58938 sb-in-f132:https ESTABLISHED
TCP 172.16.43.123:58945 cm-in-f139:https TIME_WAIT
TCP 172.16.43.123:58950 lcboma-aw-in-f5:https TIME_WAIT
TCP 172.16.43.123:58951 fx-in-f113:https ESTABLISHED
TCP 172.16.43.123:58953 lcbomp-in-f94:https TIME_WAIT
TCP 172.16.43.123:58960 lcbomo-in-f132:https ESTABLISHED
TCP 172.16.43.123:58961 sb-in-f119:https TIME_WAIT
TCP 172.16.43.123:58962 yucmhqa-in-f94:https TIME_WAIT
TCP 172.16.43.123:58963 yucmhqa-in-f94:https TIME_WAIT
TCP 172.16.43.123:58964 lcboma-au-in-f14:https ESTABLISHED
TCP 172.16.43.123:58967 pt-in-f139:https TIME_WAIT
TCP 172.16.43.123:58971 pt-in-f138:https ESTABLISHED
TCP 172.16.43.123:58972 pt-in-f113:https TIME_WAIT
TCP 172.16.43.123:58976 ø23-206-173-40:https CLOSE_WAIT
TCP 172.16.43.123:58978 150.171.85.254:https CLOSE_WAIT
TCP 172.16.43.123:58980 20.28.42.60:https CLOSE_WAIT
TCP 172.16.43.123:58987 pt-in-f132:https ESTABLISHED
TCP 172.16.43.123:58988 lcbomp-in-f102:https ESTABLISHED
TCP 172.16.43.123:58991 40.79.163.154:https ESTABLISHED

C:\Users\User>
```

Conclusion:-

The experiment was successfully performed using basic networking commands in Windows and Linux. These commands helped in viewing network configurations, testing connectivity, tracing packet routes, resolving domain names, and monitoring network connections. Such utilities are essential for network administration, troubleshooting, and maintaining reliable communication between devices.